

P19: Numerical Methods

1. Locating Roots — Change of Sign Method

Change of sign principle: If $f(a)$ and $f(b)$ have opposite signs (i.e. $f(a) \cdot f(b) < 0$) and f is continuous on $[a, b]$, then by the Intermediate Value Theorem there is at least one root in (a, b) .

Procedure

1. Evaluate $f(a)$ and $f(b)$
2. If signs differ, state: "There is a sign change in $[a, b]$, so by the IVT there is a root in this interval."
3. To narrow down: evaluate f at the midpoint and repeat (bisection)

Example: Show that $f(x) = x^3 - x - 1$ has a root between 1 and 2.

$$f(1) = 1 - 1 - 1 = -1 < 0$$

$$f(2) = 8 - 2 - 1 = 5 > 0$$

Since f is continuous and there is a sign change, there is a root in $(1, 2)$.

Limitations of the change of sign method

- **Repeated roots:** $f(x) = (x - 2)^2$ touches zero but does not change sign — no sign change detected even though $x = 2$ is a root.
- **Discontinuities:** If f has a vertical asymptote in $[a, b]$, a sign change may be detected even with no root (e.g. $f(x) = \tan x$ near $x = \pi/2$).
- **Multiple roots in interval:** The method only guarantees at least one root — there could be more (e.g. two roots, net sign unchanged).

Exam tip: Always state "sign change and f is continuous on $[a, b]$ " when justifying a root. Do not just say "one positive, one negative" — the continuity argument is required for full marks.

2. Fixed-Point Iteration

To solve $f(x) = 0$, rearrange to the form $x = g(x)$ and iterate:

$$x_{n+1} = g(x_n)$$

Starting from an initial estimate x_0 , compute $x_1, x_2, x_3, \dots$ until convergence.

Choosing a rearrangement

Different rearrangements of the same equation give different iterative formulae. Some converge; others diverge. A rearrangement $x = g(x)$ converges near a root α if:

$$|g'(\alpha)| < 1$$

If $|g'(\alpha)| > 1$, the iteration diverges.

Example: Solve $x^3 - x - 1 = 0$ using $x_{n+1} = (x_n + 1)^{1/3}$, starting at $x_0 = 1.3$.

$$x_1 = (1.3 + 1)^{1/3} = (2.3)^{1/3} \approx 1.3200$$

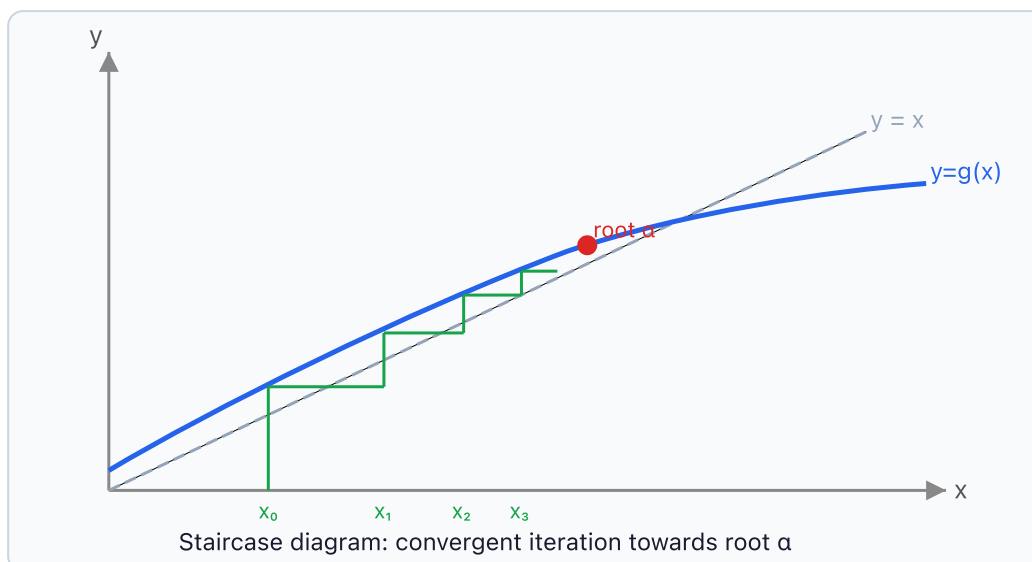
$$x_2 = (1.3200 + 1)^{1/3} \approx 1.3238$$

$$x_3 \approx 1.3245$$

Converging to root ≈ 1.3247 .

Staircase and Cobweb Diagrams

The iteration can be visualised by plotting $y = g(x)$ and $y = x$. Starting from x_0 , draw vertical and horizontal lines alternately between the two curves.



A **staircase diagram** arises when $0 < g'(\alpha) < 1$ (monotone convergence). A **cobweb diagram** (oscillating inward spiral) arises when $-1 < g'(\alpha) < 0$.

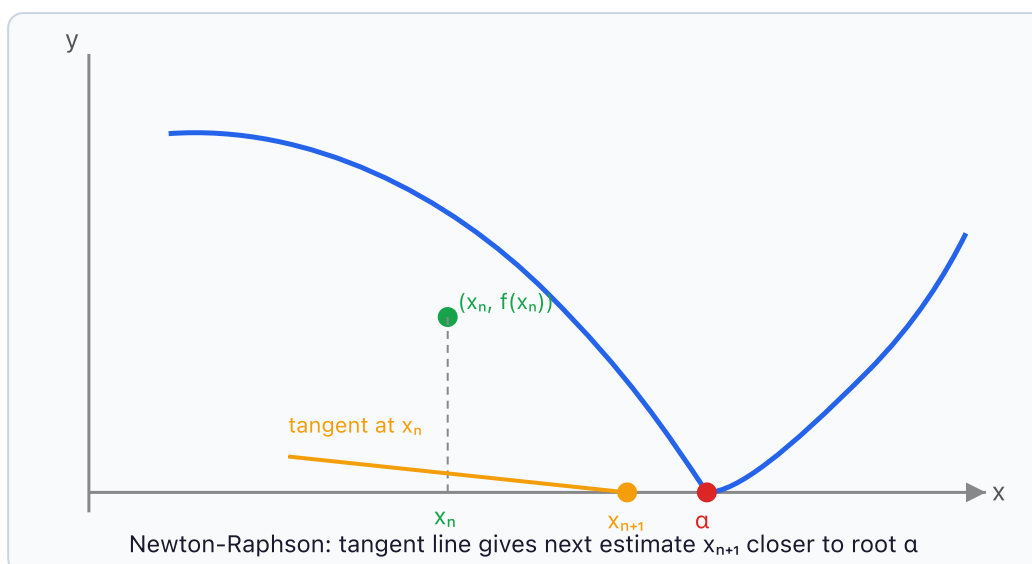
3. Newton-Raphson Method

Newton-Raphson formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This is a special case of fixed-point iteration with $g(x) = x - \frac{f(x)}{f'(x)}$. It converges much faster than basic iteration when it works.

Geometrically: at x_n , draw the tangent to $y = f(x)$. The next estimate x_{n+1} is where this tangent crosses the x -axis.



Example: Use Newton-Raphson to find the root of $f(x) = x^3 - 2x - 5$, starting at $x_0 = 2$.

$$f'(x) = 3x^2 - 2$$

$$x_1 = 2 - \frac{f(2)}{f'(2)} = 2 - \frac{8 - 4 - 5}{12 - 2} = 2 - \frac{-1}{10} = 2.1$$

$$f(2.1) = 9.261 - 4.2 - 5 = 0.061$$

$$x_2 = 2.1 - \frac{0.061}{f'(2.1)} = 2.1 - \frac{0.061}{11.23} \approx 2.0946$$

The root is approximately 2.0946.

Failure of Newton-Raphson

Newton-Raphson **fails or behaves poorly** when:

- The initial estimate x_0 is near a **stationary point** ($f'(x_0) \approx 0$) — the tangent is nearly horizontal and x_1 is far from the root.
- The function has **oscillatory behaviour** — the iterates may cycle without converging.
- The starting point is on the **wrong side of a discontinuity**.

Exam tip: Questions often ask you to explain why Newton-Raphson fails for a given starting point. The answer is almost always: "The gradient $f'(x_0)$ is very small (close to zero) because x_0 is near a stationary point, so the tangent is nearly horizontal and gives a very poor next estimate."

4. Trapezium Rule (Numerical Integration)

Approximates $\int_a^b f(x) dx$ by dividing the area into trapezoids of equal width $h = \frac{b-a}{n}$:

$$\int_a^b f(x) dx \approx \frac{h}{2} [y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n]$$

where $h = \frac{b-a}{n}$, $y_i = f(a + ih)$.

Example: Estimate $\int_0^1 e^{x^2} dx$ using 4 strips.

$h = 0.25$, x -values: 0, 0.25, 0.5, 0.75, 1

$y_0 = e^0 = 1$, $y_1 = e^{0.0625} \approx 1.0645$, $y_2 = e^{0.25} \approx 1.2840$, $y_3 = e^{0.5625} \approx 1.7551$, $y_4 = e^1 \approx 2.7183$

$$I \approx \frac{0.25}{2} [1 + 2(1.0645 + 1.2840 + 1.7551) + 2.7183] = 0.125 [1 + 2(4.1036) + 2.7183] = 0.125 \times 11.9255 \approx 1.491$$

Overestimate or underestimate?

- If f is **convex** (concave up, $f'' > 0$): trapezium rule **overestimates**
- If f is **concave** (concave down, $f'' < 0$): trapezium rule **underestimates**
- More strips = smaller h = better approximation

5. Error in Numerical Methods

To verify a root α lies in an interval, show a sign change in f over that interval at the required decimal place accuracy. For example, to confirm $\alpha = 1.325$ to 3 d.p., show $f(1.3245) < 0$ and $f(1.3255) > 0$ (or vice versa).

6. Summary Comparison

Method	Formula	Speed	When it fails
Change of sign	Sign change in f	Slow (bisection)	Repeated roots, discontinuities
Fixed-point iteration	$x_{n+1} = g(x_n)$	Moderate	$ g'(\alpha) \geq 1$
Newton-Raphson	$x_{n+1} = x_n - f/f'$	Fast (quadratic)	$f'(x_0) \approx 0$
Trapezium rule	$\frac{h}{2}[y_0 + 2y_1 + \dots + y_n]$	N/A	Inaccurate for small n